

**CLASSROOM CONTACT PROGRAMME**
(Academic Session : 2023-2024)

NEET(UG)

10-09-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEF-I (B)

This Booklet contains 24 pages.

ASSIGNMENT
(FOR SELF EVALUATION)

Name of the Student (in Capitals) : _____

Form Number : in figures _____

: in words _____

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SECTION-A (PHYSICS)

1. In a vernier callipers, one main scale division is x cm and n divisions of the vernier scale coincide with $(n - 1)$ divisions of the main scale. The least count (in cm) of the callipers is :

(1) $\left(\frac{n-1}{n}\right)x$ (2) $\frac{nx}{(n-1)}$

(3) $\frac{x}{n}$ (4) $\frac{x}{(n-1)}$

2. A physical quantity $P = \frac{Q^2 R^{1/2}}{S}$ where Q , R , S are different physical quantities. If error in measurement of Q is 4%, R is 2% and S is 3% then find percentage error in measurement of P :-

- (1) 6 % (2) 12 % (3) 9 % (4) 13 %

3. The numbers 2.745 and 2.735 on rounding off to 3 significant figures will be :

- (1) 2.75 and 2.74 (2) 2.74 and 2.73

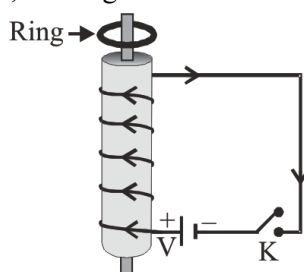
- (3) 2.75 and 2.73 (4) 2.74 and 2.74

4. If momentum (P), area (A) and time (T) are taken to be fundamental quantities, then energy has the dimensional formula :

(1) $[P^1 A^{-1} T^1]$ (2) $[P^2 A^1 T^1]$

(3) $[P^1 A^{-1/2} T^1]$ (4) $[P^1 A^{1/2} T^{-1}]$

5. A conducting ring is placed around the core of an electromagnet as shown in figure. When key K is pressed, the ring.



- (1) Remain stationary
(2) Is attracted towards the electromagnet
(3) Jumps out of the core
(4) None of the above

6. **Assertion (A)** :- Only a change of magnetic flux with time, will maintain an induced current in the coil.

Reason (R) :- The presence of a large magnetic flux will maintain an induced current in the coil.

- (1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

- (2) **(A)** is correct but **(R)** is not correct

- (3) **(A)** is incorrect but **(R)** is correct

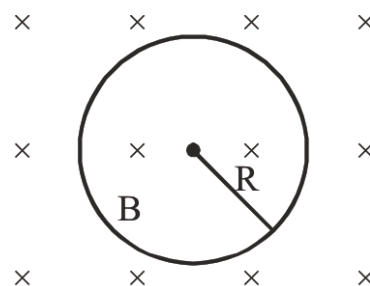
- (4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

7. A coil having effective area A , is held with its plane normal to a magnetic field of induction B . The magnetic induction is quickly reduced to 25% of its initial value in 2s. Then e.m.f induced across the coil will be

(1) $\frac{3AB}{8}$ (2) $\frac{3AB}{4}$

(3) $\frac{AB}{4}$ (4) $\frac{AB}{2}$

8. A conductor loop of radius R is present in a uniform magnetic field B perpendicular to the plane of the ring. If radius R varies as a function of time ' t ' as $R = R_0 + t$. The e.m.f. induced in the loop and direction of induced current is:-



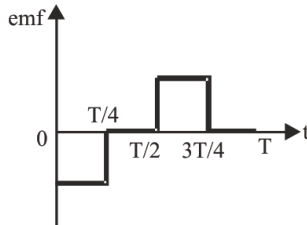
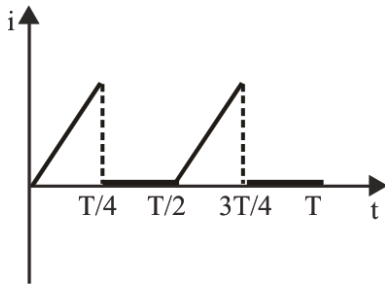
- (1) $2\pi(R_0 + t)B$, clockwise
(2) $\pi(R_0 + t)B$, clockwise
(3) $2\pi(R_0 + t)B$, anticlockwise
(4) Zero

9. Two solenoids have identical geometrical construction and same number of turns but one is made of thick wire and other of thin wire. Which of the following quantities are different for two solenoids -

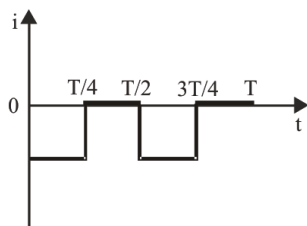
(a) Self inductance.
 (b) Rate of joule heating if the same current goes through them.
 (c) Magnetic potential energy if the same current goes through them.
 (d) Time constant.

- (1) b, d (2) a, c
 (3) b, c (4) All above

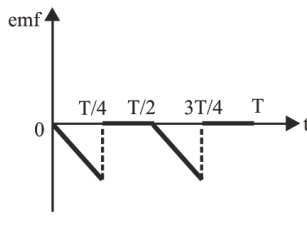
10. The current i in a coil varies with time as shown in the figure. The variation of induced emf with time would be :



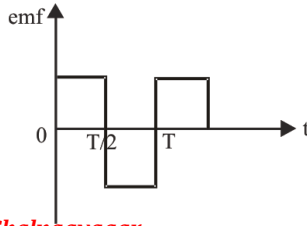
- (1) 0



- (2) 0



- (3) 0

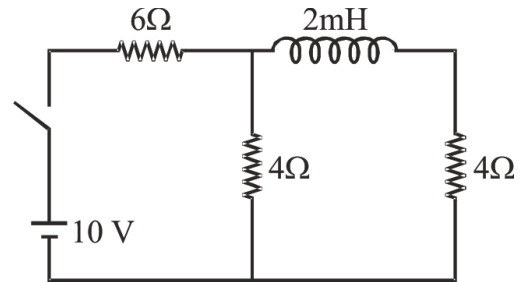


- (4) 0

11. In LR circuit the current increases to one fourth of its maximum value in 4sec, then the time constant of the circuit is -

- (1) $2\log_e 2$ (2) $\frac{4}{\log_e 2}$
 (3) $\frac{4}{\log_e \left(\frac{4}{3}\right)}$ (4) $\frac{\log_e 2}{2}$

12. In the given circuit find the ratio of i_1 to i_2 . Where i_1 is the initial (at $t = 0$) current, and i_2 is steady state (at $t = \infty$) current through the battery :-

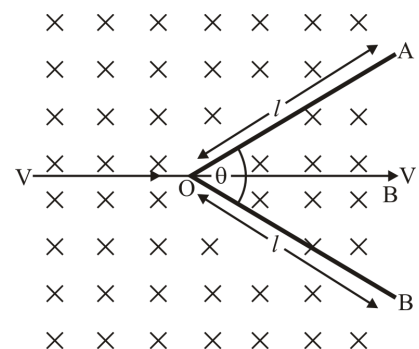


- (1) 1.0 (2) 0.8 (3) 1.2 (4) 1.5

13. A current $I = 10\sin(100\pi t)$ A is passed in first coil, which induces a maximum emf of 5π V in second coil. The mutual inductance between the coil is :-

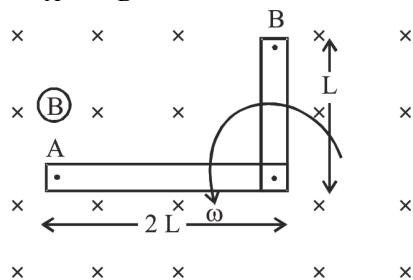
- (1) 5 mH (2) 10 mH
 (3) 15 mH (4) 20 mH

14. A conductor AOB with angle $\angle AOB = \theta$ moves with velocity v in a magnetic field B as shown in Fig. The emf induced is :-



- (1) $2Blv \sin \theta$
 (2) $2Blv \cos \theta$
 (3) $2Blv \sin \theta/2$
 (4) zero

15. Find out $V_A - V_B = ?$



- (1) $B\omega (2L)^2$ (2) $\frac{B\omega (2L)^2}{2}$
 (3) $\frac{3}{2}B\omega L^2$ (4) $-\frac{3B\omega L^2}{2}$

16. A coil of 20×20 cm having 30 turns is rotating 30 rps in a magnetic field of induction 1 tesla. The peak value of the induced emf is approximately:-

- (1) 452 volt
 (2) 226 volt
 (3) 113 volt
 (4) 339 volt

17. In steady state condition for following cases, match correct graph between temperature and distance ($T_1 \rightarrow$ Temperature of left end is high and x is distance measured from left end) :-

(A)		(P)	
(B)		(Q)	
(C)		(R)	
		(S)	None of these

- (1) $A \rightarrow Q, B \rightarrow R, C \rightarrow S$
 (2) $A \rightarrow P, B \rightarrow R, C \rightarrow Q$
 (3) $A \rightarrow R, B \rightarrow P, C \rightarrow S$
 (4) $A \rightarrow S, B \rightarrow Q, C \rightarrow R$

18. A liquid cools down from 70°C to 60°C in 5 minutes. The time taken to cool it from 60°C to 50°C will be :-

- (1) 5 minutes
 (2) Lesser than 5 minutes
 (3) Greater than 5 minutes
 (4) Lesser or greater than 5 minutes depending upon the density of the liquid

19. The adiabatic Bulk modulus of a diatomic gas at atmospheric pressure is

- (1) 0 Nm^{-2}
 (2) 1 Nm^{-2}
 (3) $1.4 \times 10^4 \text{ Nm}^{-2}$
 (4) $1.4 \times 10^5 \text{ Nm}^{-2}$

20. A refrigerator works between temperature -10°C and 27°C , the coefficient of performance is :

- (1) 7.1 (2) 1
 (3) 8.1 (4) 15.47

21. Heat is supplied to a diatomic gas at constant pressure. The ratio of $\Delta Q : \Delta U : \Delta W$ is :-

- (1) 5 : 3 : 2
 (2) 5 : 2 : 3
 (3) 7 : 5 : 2
 (4) 7 : 2 : 5

22. A Carnot engine absorbs an amount Q of heat from a reservoir at an absolute temperature T and rejects heat to a sink at a temperature of $T/3$. The amount of heat rejected is :-

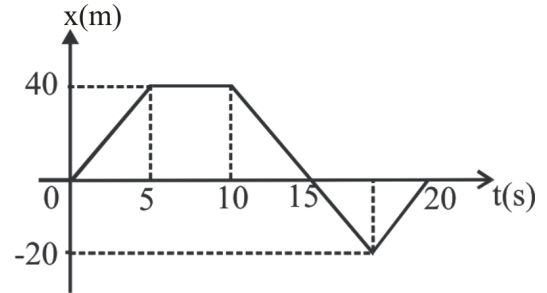
- (1) $Q/4$ (2) $Q/3$
 (3) $Q/2$ (4) $2Q/3$

23. The angle of polarisation for any medium is 60° , what will be critical angle for this -

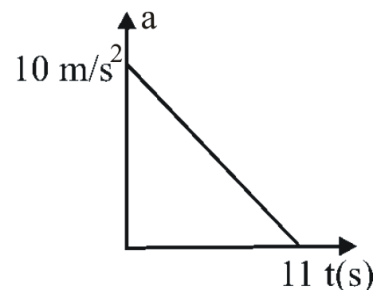
- (1) $\sin^{-1}\sqrt{3}$ (2) $\tan^{-1}\sqrt{3}$
 (3) $\cos^{-1}\sqrt{3}$ (4) $\sin^{-1}\frac{1}{\sqrt{3}}$

24. Wavelength of light used in an optical instrument are $\lambda_1 = 4000 \text{ \AA}$ and $\lambda_2 = 5000 \text{ \AA}$, then ratio of their respective resolving powers (corresponding to λ_1 and λ_2) is-
- (1) 16 : 25 (2) 9 : 1 (3) 4 : 5 (4) 5 : 4
25. A slit of width d is placed in front of a screen at a distance of 0.5 m and is illuminated normally with light of wavelength $5.89 \times 10^{-7} \text{ m}$. The first diffraction minima on either side of central diffraction maximum are separated by $2 \times 10^{-3} \text{ m}$. Find the width of the slit :-
- (1) $2.95 \times 10^{-4} \text{ m}$ (2) $5.45 \times 10^{-4} \text{ m}$
 (3) $9.1 \times 10^{-4} \text{ m}$ (4) $1.47 \times 10^{-4} \text{ m}$
26. Determine the angular separation between central maximum and first order maxima of the diffraction pattern due to a single slit of width 0.25 mm when light of wavelength $5890 \text{ \AA}$ is incident on it :-
- (1) $3.534 \times 10^{-3} \text{ rad}$
 (2) $2.35 \times 10^{-3} \text{ rad}$
 (3) $1.47 \times 10^{-4} \text{ rad}$
 (4) $15.8 \times 10^{-3} \text{ rad}$
27. In Young's double slit experiment, the fringes are displaced by a distance x when a glass plate of refractive index 1.5 is introduced in the path of one of the beams. When this plate is replaced by another plate of the same thickness, the shift of fringes is $(3/2)x$. The refractive index of the second plate is :-
- (1) 1.75 (2) 1.50 (3) 1.25 (4) 1.00
28. In Young's double slit experiment, if monochromatic light is replaced by white light :
- (1) All bright fringes become white
 (2) All bright fringes have colours between violet and red
 (3) Only the central fringe is white, all other fringes are coloured
 (4) No fringes are observed

29. Two coherent monochromatic light waves of intensities I and $9I$ are superposed. The maximum and minimum possible intensities in the resulting beam are -
- (1) $5I$ and I (2) $16I$ and $4I$
 (3) $9I$ and I (4) $9I$ and $3I$
30. A boy begins to walk eastward along a street in front of his house and his position-time graph is shown below. The average speed of boy during entire journey is :-



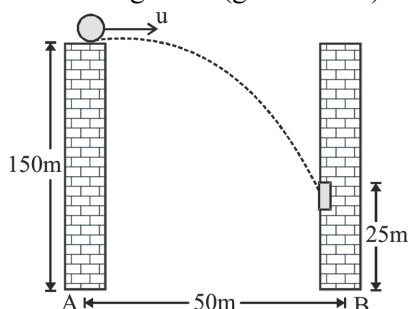
- (1) 8 m/s (2) 6 m/s
 (3) $\frac{8}{3} \text{ m/s}$ (4) 2 m/s
31. A particle starts from rest. Its acceleration (a) versus time (t) is as shown in the figure. The maximum speed of the particle will be :-



- (1) 110 m/s (2) 55 m/s
 (3) 550 m/s (4) 660 m/s
32. The point from where a ball is projected is taken as the origin of the co-ordinate axes. The x and y components of its displacement are given by $x = 6t$ and $y = 8t - 5t^2$. What is the velocity of projection ?
- (1) 6 ms^{-1}
 (2) 8 ms^{-1}
 (3) 10 ms^{-1}
 (4) 14 ms^{-1}

33. Particle is projected upward with velocity 100 m/s at an angle of 30° with horizontal. Find the time when its velocity is perpendicular to the acceleration :-
 (1) 10 s (2) 5 s (3) 20 s (4) 15 s

34. Two buildings A and B are 50m apart as shown, with what horizontal velocity 'u' a ball must be thrown so that it enters into window on building B, 25m above the ground ($g = 10 \text{ m/s}^2$):-



- (1) 10 m/s (2) 15 m/s
 (3) 20 m/s (4) 8 m/s
35. A particle projected from origin follows the trajectory $y = 3x - \frac{x^2}{9}$, where x-axis is along horizontal direction and y-axis is along vertical direction. Find the horizontal range of projectile :-
 (1) 27 m (2) 9 m
 (3) $3\sqrt{3}$ m (4) $9\sqrt{3}$ m

SECTION-B (PHYSICS)

36. Two full turns of Screw gauge cover a distance of 1 mm on main scale. The total number of circular scale divisions are 100. The measured value of thickness of sheet will be:

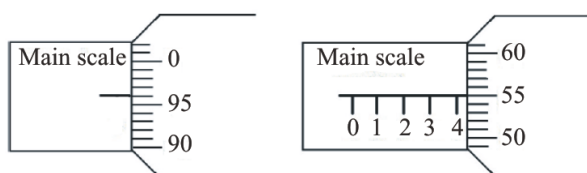


Fig.(1) SCREW GAUGE Fig.(2)

- (1) 4.555 mm
 (2) 4.295 mm
 (3) 2.295 mm
 (4) 2.245 mm

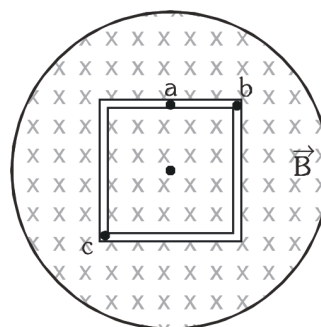
37. Two identical circular loops of metal wire are lying on a table without touching each other. Loop A carries a current, which increases with time. In response, loop B :-

- (1) remains stationary
 (2) is attracted by the loop A
 (3) is repelled by the loop A
 (4) rotates about its CM, with CM fixed

38. A time varying magnetic flux passing through a coil is given by $\phi = xt^2$, if at $t = 3\text{s}$, the emf induced is 9V, then the value of x is

- (1) 0.66 Wb/s^2 (2) -0.66 Wb/s^2
 (3) 1.5 Wb/s^2 (4) -1.5 Wb/s^2

39. A square conducting loop is placed in the time varying magnetic field $\left(\frac{dB}{dt} = +\text{ve constant}\right)$. The centre of square coincides with axis of cylindrical region of magnetic field. The directions of induced electric field at point a, b and c.



- (1) (2) (3) (4)

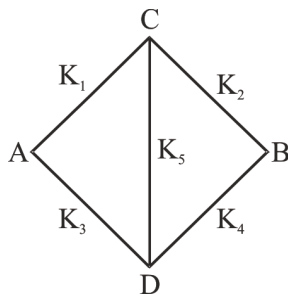
40. A horizontal straight conductor of length L placed along east-west direction falls under gravity from height h at a place where horizontal component is H and vertical component is V . Then :-

- (1) no emf is induced along the length of the conductor
- (2) an emf is induced along the length which has maximum value $HL\sqrt{2gh}$
- (3) an emf is induced along the length which has maximum value $VL\sqrt{2gh}$
- (4) an emf is induced along the length which has maximum value $L\sqrt{2gh(V^2 + H^2)}$

41. If the rotational velocity of a dynamo armature is doubled, then induced e.m.f will become :-

- (1) half
- (2) two times
- (3) four times
- (4) unchanged

42. Five rods of same dimensions are arranged as shown in figure. They have thermal conductivities K_1, K_2, K_3, K_4 and K_5 . When points A and B are maintained at different temperatures. No heat flows through the central rod if :



- (1) $K_1 K_4 = K_2 K_3$
- (2) $K_1 K_2^2 = K_3 K_4^2$
- (3) $\frac{K_1}{K_4} = \frac{K_2}{K_3}$
- (4) $K_1 K_2 = K_3 K_4$

TG: @Chalnaayaaar

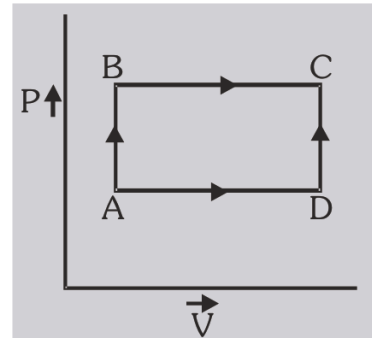
PHASE - MEF

43. **Assertion (A)** :- According to Newton's law of cooling, the rate of loss of heat $-dQ/dt$ of the body is directly proportional to the difference of temperature between body and surrounding.

Reason (R) :- This law holds for all type of temperature differences.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is incorrect but (R) is correct

44. As shown in the figure the amount of heat absorbed along the path ABC is 90J and the amount of work done by the system is 30 J. If the amount of work done along the path ADC is 20 J then amount of heat absorbed will be :-



- (1) 80 J
- (2) 90 J
- (3) 110 J
- (4) 120 J

45. In YDSE, $d = 1 \text{ mm}$, $\lambda = 6000 \text{ \AA}$ and $D = 1 \text{ m}$. The minimum distance between two points on screen having 75% intensity of the maximum intensity will be :-

- (1) 0.50 mm
- (2) 0.40 mm
- (3) 0.30 mm
- (4) 0.20 mm

46. Young's Double Slit Experiment is first performed in air then in a medium other than air. It is found that 10th bright fringe in medium lies where 8th dark fringe lies in air. The refractive index of medium is nearly :-

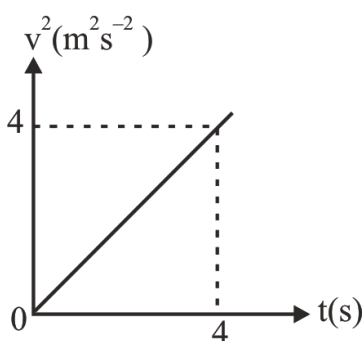
- (1) 1.59
- (2) 1.78
- (3) 1.69
- (4) 1.33

47. **Assertion (A)** :- For best contrast between maxima and minima in the interference pattern of Young's double slit experiment, the intensity of light emerging out of the two slits should equal.

Reason (R) :- The intensity of interference pattern is directly proportional to the square of amplitude.

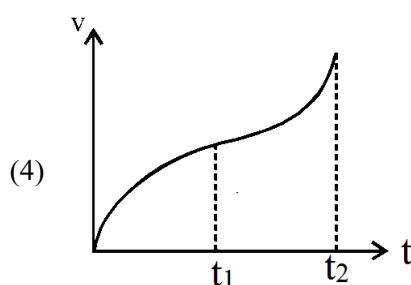
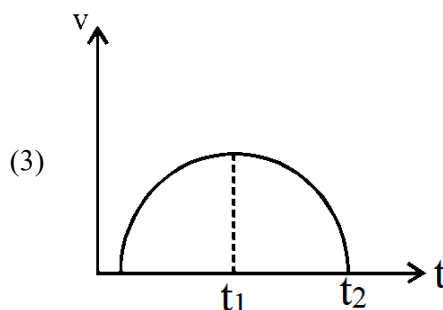
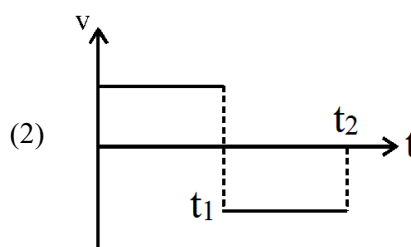
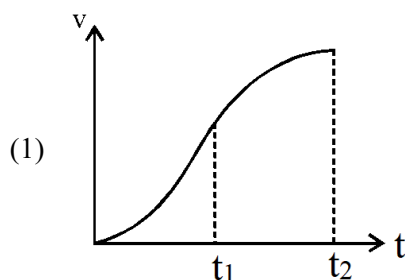
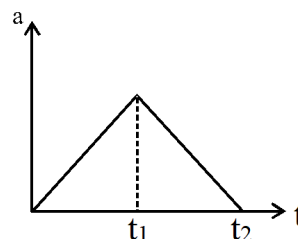
- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (2) (A) is correct but (R) is not correct
- (3) (A) is incorrect but (R) is correct
- (4) Both (A) and (R) are correct but (R) is the correct explanation of (A)

48. A particle is moving along a straight line such that square of its velocity varies with time as shown in the figure. What is the acceleration of the particle at $t = 4$ s ?



- (1) 4 m/s²
- (2) 1/4 m/s²
- (3) 1/2 m/s²
- (4) 0

49. The acceleration-time graph of a particle is shown. The respective velocity-time graph of the particle is:



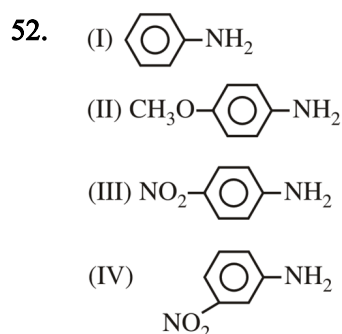
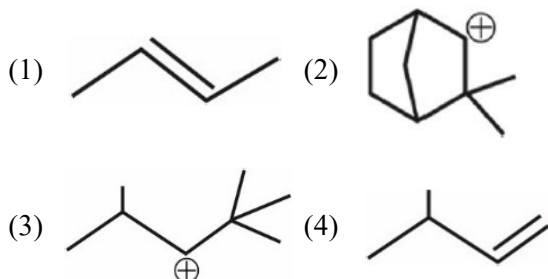
50. A ball is projected upwards from the top of tower with a velocity 50 ms⁻¹ making an angle 30° with the horizontal. The height of tower is 70 m. After how many seconds from the instant of throwing will the ball reach the ground :-

- (1) 2s
- (2) 5s
- (3) 7s
- (4) 9s

Topic : IONIC EQUILIBRIUM, THERMODYNAMICS (TILL FLOT), CHEMICAL BONDING (IONIC BOND ONWARDS), s-BLOCK, COORDINATION COMPOUND [BACK UNIT] : GOC-I

SECTION-A (CHEMISTRY)

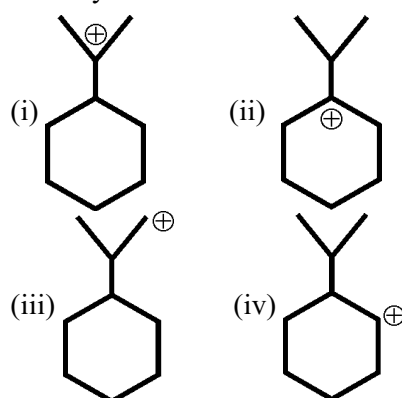
51. Hyperconjugation is not possible in



The correct order of decreasing basicity of the above compounds is.

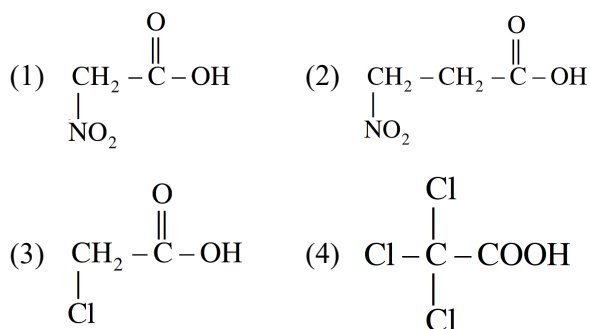
- (1) I > II > III > IV (2) II > I > IV > III
 (3) III > IV > II > I (4) II > I > III > IV

53. Arrange the following carbocation in order of stability



- (1) i > ii > iii > iv
 (2) i > ii > iv > iii
 (3) ii > i > iv > iii
 (4) ii > i > iii > iv

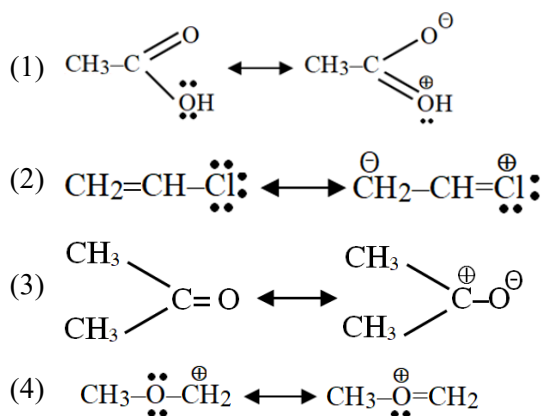
54. Which of the following acids has lowest pK_a value?



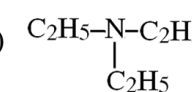
55. Correct order of -I effect of the given species is:

- (1) $-\text{CN} > \text{NH}_3^+ > -\text{F}$
 (2) $\text{NH}_3^+ > -\text{CN} > -\text{F}$
 (3) $\text{NH}_3^+ > -\text{F} > -\text{CN}$
 (4) $-\text{F} > \text{NH}_3^+ > -\text{CN}$

56. In which of the following pairs, structure on the right is a major resonance contributor ?



57. Most basic compound among following is -

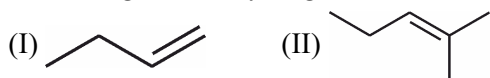
- (1) $\text{C}_2\text{H}_5 - \text{NH}_2$
 (2) 
 (3) $\text{C}_2\text{H}_5 - \text{NH} - \text{C}_2\text{H}_5$
 (4) NH_3

58. **Assertion** : The pK_a of acetic acid is lower than that of phenol.

Reason : Phenoxide ion is more resonance stabilized than acetate ion.

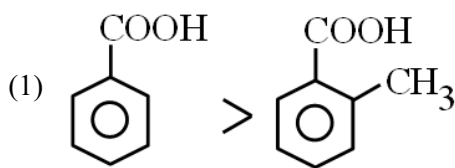
- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.

59. Rank the following alkenes in order of decreasing heat of hydrogenation

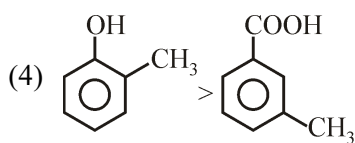
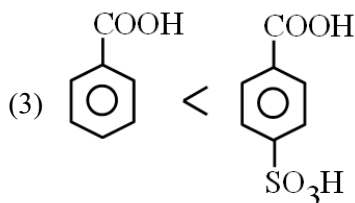


- (1) II > III > IV > I
- (2) II > IV > III > I
- (3) I > III > IV > II
- (4) I > IV > III > II

60. Correct order of acidic strength ?



(2) Ph-COOH > H-COOH



61. Which of the following halide cannot be converted into anhydrous halide on direct heating ?

- (1) $BaCl_2 \cdot 6H_2O$ (2) $CaCl_2 \cdot 6H_2O$
- (3) $BeCl_2 \cdot 6H_2O$ (4) $SrCl_2 \cdot 6H_2O$

62. Identify incorrect match of colour in flame test.

- (1) Li - Crimson red
- (2) Na - Golden yellow
- (3) K - Blue
- (4) Rb - Red-violet

63. Alkali metal solution in liquid NH_3 , blue colour fade and become colourless, when

- (1) more alkali metal added
- (2) more NH_3 is added
- (3) more ammoniated electron form
- (4) sodium amide form with H_2 gas

64. Sodium losses its lustre on exposure to moist air due to formation of

- (1) Na_2O , $NaOH$ and Na_2CO_3
- (2) Na_2O and $NaOH$
- (3) Na_2O and Na_2CO_3
- (4) $NaOH$ and Na_2CO_3

65. The correct order of density of following elements is

Be, Mg, Ca, Sr

- (1) Be > Mg > Ca > Sr
- (2) Ca > Mg > Be > Sr
- (3) Ca < Mg < Be < Sr
- (4) Mg < Ca < Sr < Be

66. EAN value of complex $[Ni(en)_2]^{+2}$ ($z = 28$)

- (1) 36 (2) 34
- (3) 38 (4) 30

67. In $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$, sodium nitroprusside :-

- (1) Oxidation state of Fe is +2
- (2) It has NO^+ as ligand
- (3) Both 1 & 2 are correct
- (4) Coordination number = 4

68. Which of the following ligands will not show chelation?

- (1) EDTA
- (2) DMG
- (3) Ethane-1, 2 diamine
- (4) SCN^-

69. The IUPAC nomenclature for the complex $\text{Na}[\text{PtBrCl}(\text{NO}_2)(\text{NH}_3)]$ is :-

- (1) Sodium amminechloridobromidonitrito-N-platinum(II)
- (2) Sodium nitritochloridobromidoammine-N-platinate (II)
- (3) Sodium amminebromidochloridonitrito-N-platinate (II)
- (4) Sodium amminebromidochloridonitrito-N-platinate (III)

70. Given the IUPAC name of the complex compound $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$:-

- (1) Bromoaquatetraamine Cobalt (III) nitrate
- (2) Bromoaquatetraaminocobalt (III) nitrate
- (3) Bromoaquaterraamine cobalt (III) nitrate
- (4) Tetraammineaquabromido cobalt (III) nitrate

71. CN^- is a :-

- (1) mono dentate ligands
- (2) ambidentate ligands
- (3) π -acid ligands
- (4) All of these

72. **Statement-1** :- Na_2CO_3 can be manufactured by solvay's process but K_2CO_3 can not be prepared.

Statement-2 :- KHCO_3 is more soluble than NaHCO_3 .

- (1) Statement-1 & 2 both are correct
- (2) Statement-1 & 2 both are incorrect
- (3) Statement-1 is correct but statement-2 is incorrect
- (4) Statement-1 is incorrect but statement-2 is correct

73. At 80°C , Pure distilled water has H_3O^+ concentration equal to $1 \times 10^{-6} \text{ mol L}^{-1}$. The value of K_w at this temperature will be :-

- (1) 1×10^{-8}
- (2) 1×10^{-12}
- (3) 1×10^{-14}
- (4) 10^{-6}

74. Solubility of Zirconium phosphate $\text{Zr}_3(\text{PO}_4)_4$ is 'S' moles per litre. Solubility product K_{sp} may be given as

- (1) $6912 S^7$
- (2) $408 S^7$
- (3) $4S^3$
- (4) None

75. Precipitation occurs only if I.P (Ionic Product)

- (1) Equals K_{sp}
- (2) Exceeds K_{sp}
- (3) Less than K_{sp}
- (4) Is very small

76. Which among the following can act both as Bronsted acid as well as a Bronsted base?

- (1) CH_3COO^-
- (2) HSO_4^-
- (3) Cl^-
- (4) NH_4^+

77. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The pH of dimethylammonium acetate solution is :

- (1) 8.50
- (2) 5.50
- (3) 7.75
- (4) 6.25

78. The solubility products of three sparingly soluble salts are given below :

No. Formula Solubility product

- | | | |
|---|---------------|-----------------------|
| 1 | PQ | 4.0×10^{-20} |
| 2 | PQ_2 | 3.2×10^{-14} |
| 3 | PQ_3 | 2.7×10^{-35} |

The correct order of decreasing molar solubility is :-

- (1) 1, 2, 3
- (2) 2, 1, 3
- (3) 3, 2, 1
- (4) 2, 3, 1

79. Find the solubility of Na_2SO_4 ($K_{sp} = 4 \times 10^{-15}$) in the presence of 10^{-3} M CaSO_4 solution :-

- (1) 10^{-12} (2) 4×10^{-6}
(3) 10^{-6} (4) 4×10^{-12}

80. Which of the following statement(s) is correct for weak electrolyte.

- (1) Degree of ionisation (α) increases with increase in volume of container according to Ostwald's dilution law.
(2) Degree of ionisation (α) decreases with increases in pressure.
(3) Degree of ionisation (α) increases with decreases in concentration
(4) (1) & (3) both

81. Which amongst the following is strongest conjugate base

- (1) HSO_4^- (2) ClO_4^- (3) NO_3^- (4) H_2O

82. What is the percentage ionisation of 0.01 M HA solution ($K_a = 10^{-6}$)

- (1) 9.5 % (2) 1 %
(3) 10.5 % (4) 17 %

83. Upon hydrolysis of sodium carbonate, the reaction takes place between

- (1) Na^+ & Water (2) Na^+ & OH^-
(3) CO_3^{2-} & Water (4) CO_3^{2-} & H^+

84. Which of the following is not state function.

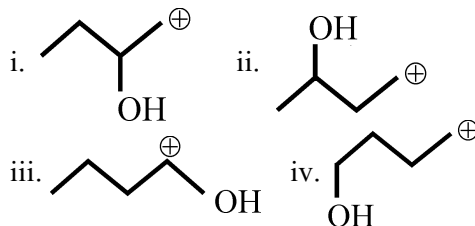
- (1) Internal energy
(2) Enthalpy
(3) Work
(4) Entropy

85. Among them intensive property is

- (1) Mass (2) Volume
(3) Surface tension (4) Enthalpy

SECTION-B (CHEMISTRY)

86. Compare stability order of given carbocation.



- (1) $i > ii > iii > iv$ (2) $iv > ii > i > iii$
(3) $iii > iv > ii > i$ (4) $iii > iv > i > ii$

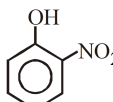
87. Which of the following is not a nucleophile ?

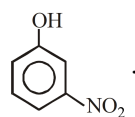
- (1) OH^- (2) NH_3
(3) BF_3 (4) $\text{C}_2\text{H}_5\text{OH}$

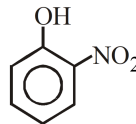
88. **Assertion :-** Pyridine is more basic than aniline.

Reason :- Pyridine have resonance.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

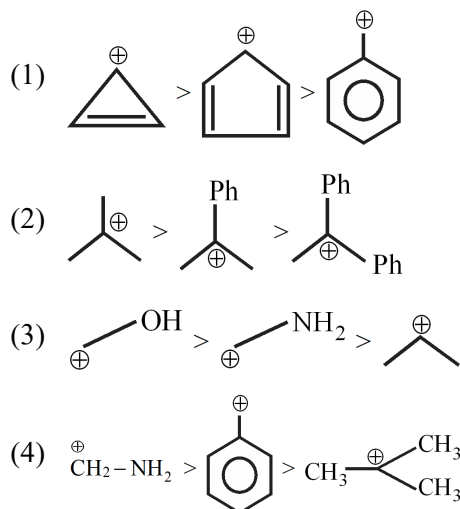
89. **Assertion :-**  is more acidic than



Reason :- Intra molecular H-bond in  is present.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

90. Which of the following order is correct for stability of carbocation.



91. Which of following complex will give 2 mole AgCl precipitate when react with excess AgNO₃.

- (1) PtCl₄.2NH₃
(2) PtCl₂.4NH₃
(3) PtCl₄.3NH₃
(4) PtCl₄.6NH₃

92. An oxide of alkaline earth metals (X) react with Cl₂ to give a compound (Y). Y is found in polymer chain structure and is electron deficient molecule. The compound Y is :

- (1) BeO
(2) BeCl₂
(3) Be(OH)₂
(4) BeCO₃

93. Which of the following carbonyls will have the strongest C–O bond ?

- (1) Fe(CO)₅ (2) Mn(CO)₆⁺
(3) Cr(CO)₆ (4) V(CO)₆[–]

94. According to EAN rule, how many CO group should be attached to Fe ?

- (1) 4 (2) 5
(3) 6 (4) 8

95. Which of the following statement is correct with respect to the metal carbonyls of Ist transition series ?

- (1) As M – C, π bonding increases, the C – O bond length increases.
(2) As positive charge on the central metal atom increases, the C – O bond length increases.
(3) As electron density on the central metal atom increases, the C – O bond length increases.
(4) (1) and (3) both.

96. At 90°C, pure water has $[H^+] = 10^{-6}M$, if 100 mL of 0.2 M HCl is added to 200 mL of 0.1 M KOH at 90°C then pH of the resulting solution will be

- (1) 5 (2) 6 (3) 7 (4) 4

97. What is the percentage degree of hydrolysis (approx.) of KCN in $\frac{N}{80}$ solution when the dissociation constant for HCN is 1.6×10^{-9} and $K_w = 10^{-14}$.

- (1) 5.26 (2) 2.24 (3) 8.2 (4) 11.21

98. Amongst the following given electrolyte, salt having minimum pH.

- (1) NaCN (2) Na₂SO₄
(3) KCl (4) MgCl₂

99. The solubility of electrolytes MX₁, MX₂, MX₃, is 10^{-3} mol/l . Hence their respective solubility products are-

- (1) 10^{-6} , 4×10^{-9} , 27×10^{-12}
(2) 10^{-9} , 4×10^{-9} , 32×10^{-12}
(3) 10^{-9} , 8×10^{-8} , 32×10^{-12}
(4) 10^{-3} , 4×10^{-6} , 32×10^{-9}

100. When equal volume of AgNO₃ & NaCl solution are mixed, the precipitation of AgCl ($K_{sp} = 1.81 \times 10^{-10}$) will occur with

- (1) $10^{-3} M Ag^+$ & $10^{-10} M Cl^-$
(2) $10^{-5} M Ag^+$ & $10^{-5} M Cl^-$
(3) $10^{-6} M Ag^+$ & $10^{-5} M Cl^-$
(4) $10^{-4} M Ag^+$ & $10^{-4} M Cl^-$

Topic : MONERA TO GYMNASPERMS, MORPHOLOGY (COMPLETE), UPTO DIGESTIVE GLANDS (CHAPTER-DIGESTIVE SYSTEM), PHOTOSYNTHESIS-INTRODUCTION [BACK UNIT] : ECOLOGY

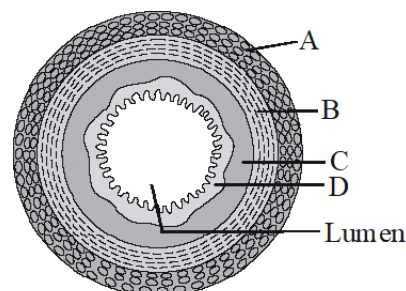
SECTION-A (BIOLOGY-I)

101.is the largest salivary gland
- (1) Submandibular (2) Sublingual
(3) Parotid (4) None
102. Select the correct match
- (1) Duodenum - Cartilaginous flap
(2) Epiglottis - Opening of wind pipe
(3) Glottis - C-shaped structure
(4) Caecum - Small blind sac
103. Oblique muscles are present in :
- (1) Stomach (2) Kidneys
(3) Intestine (4) Heart
104. Gastric juice of adult contains:-
- (1) Nuclease, pepsinogen
(2) Pepsinogen, lipase
(3) Amylase, rennin
(4) Maltase, pepsinogen
105. In stomach food remains for the
- (1) 14 to 15 hours (2) 13 to 14 hours
(3) 4 to 5 hours (4) 9 to 10 hours
106. Dentition in human is :-
- (1) Thecodont, heterodont and polyphyodont
(2) Thecodont, homodont and polyphyodont
(3) Thecodont, heterodont and diphyodont
(4) Acrodont, heterodont and diphyodont
107. Which one serves as common passage for both food and air ?
- (1) Larynx (2) Pharynx
(3) Gullet (4) Glottis

108. How many deciduous teeth are present in human ?

- (1) 22 (2) 24
(3) 20 (4) 18

109. The below diagram represents the TS of gut. Identify A, B, C and D :-



- (1) A-Serosa, B-Muscularis, C-Submucosa, D-Mucosa
(2) A-Muscularis, B-Serosa, C-Submucosa, D-Mucosa
(3) A-Serosa, B-Muscularis, C-Mucosa, D-Submucosa
(4) A-Serosa, B-Submucosa, C-Muscularis, D-Mucosa

110. Match the two columns and select the correct amongst options given below :

Column I		Column II	
A.	Biomacromolecules of food	i.	Alimentary canal and associated gland
B.	Human digestive system	ii.	Embedded in jawbones.
C.	Stomach	iii.	The outer wall of visceral organs
D.	Thecodont	iv.	Converted into simple substances
E.	Serosa	v.	J-shaped bag-like structure

- (1) A-ii, B-i, C-v, D-iii, E-iv
(2) A-iv, B-i, C-v, D-ii, E-iii
(3) A-i, B-ii, C-iii, D-iv, E-v
(4) A-i, B-iii, C-ii, D-iv, E-v

111. Match the two columns and select the right one among options given

Column I		Column II	
A.	Duodenum	i.	A cartilaginous flap
B.	Epiglottis	ii.	Small blind sac
C.	Glottis	iii.	'C' shaped structure emerging from the stomach
D.	Caecum	iv.	Opening of windpipe

- (1) A-i, B-ii, C-iii, D-iv
 (2) A-iv, B-iii, C-ii, D-i
 (3) A-iii, B-i, C-iv, D-ii
 (4) A-ii, B-iv, C-i, D-iii
112. Serosa is :-
- (1) Innermost layer of alimentary canal
 (2) Made up of thin mesothelium
 (3) Outermost layer of alimentary canal
 (4) Both (2) & (3)
113. Find out the correct match from the following table :-

	Column-I	Column-II	Column-III
A	Goblet cells	Mucus	Prevent mucosa layer from damage of HCl
B	Lysozyme	Saliva	Antibacterial agent
C	Saliva	Submaxillary gland	Secrete salivary amylase
D	Chief cells	HCl	Stimulate gastric lipase

- (1) Only A
 (2) A and B
 (3) Only C
 (4) C and D
114. Select the correct statements.
- (a) Salivary glands situated just outside the buccal cavity.
 (b) Liver is the largest digestive gland.
 (c) Hepatic duct not arise from gall bladder.
 (d) Sphincter of Oddi can regulate the release of pancreatic and bile juice in duodenum.
- (1) a, b, c & d
 (2) a, b & d
 (3) b, c & d
 (4) a, b & c

115. Read the following statements and find out the incorrect statements :-

(a) The alimentary canal begins with an anterior opening- the mouth, and it opens out posteriorly through the anus.
 (b) The mouth leads to the buccal cavity or oral cavity. The oral cavity has a number of teeth, a muscular tongue and three pairs of salivary glands.
 (c) The first set of teeth is called temporary milk or adult teeth which are replaced by a set of permanent or deciduous teeth.
 (d) The upper surface of the tongue has small projections called taste buds.

- (1) a and b
 (2) b and c
 (3) c and d
 (4) b, c and d

116. Given below are two statements:

Statement-I : In wall of alimentary canal, serosa is outermost layer made of thin mesothelium.

Statement-II : Muscularis layer in stomach has outer circular muscle and inner longitudinal muscle.

In the light of the above statements, choose the most appropriate answer from the options given below:

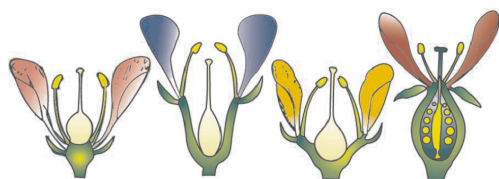
- (1) Both Statement-I and II both are incorrect.
 (2) Statement-I is correct but statement-II is incorrect.
 (3) Statement-I is incorrect but statement-II is correct.
 (4) Both Statement-I and II both are correct.

117. Match the following floral formula correctly with related family

(P)	$\oplus \frac{\uparrow}{\downarrow} K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$	(A)	Liliaceae
(Q)	$\% \frac{\uparrow}{\downarrow} K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_1$	(B)	Solanaceae
(R)	$\oplus \frac{\uparrow}{\downarrow} K_{(5)} \overset{\curvearrowright}{C}_{(5)} A_5 \underline{G}_{(2)}$	(C)	Brassicaceae
(S)	$\text{Br} \oplus \frac{\uparrow}{\downarrow} \overset{\curvearrowright}{P}_{(3+3)} A_{3+3} \underline{G}_{(3)}$	(D)	Fabaceae

- (1) P - D; Q - C, R - B, S - A
 (2) P - C; Q - D, R - B, S - A
 (3) P - C; Q - D, R - A, S - B
 (4) P - A; Q - C, R - B, S - D

118. The four sketch A,B,C given below, represent four different type flower. Which one of these is correctly identified in the option



	A	B	C
(1)	Rose	China rose	Peach
(2)	Brinjal	Mustard	Cucumber
(3)	Guava	Plum	Apple
(4)	Brinjal	Peach	Cucumber

119. Match the column I, II and III and choose the correct option.

	Column-I		Column-II		Column-III
I	Marginal	A		M	Argemone
II	Axile	B		N	Dianthus
III	Parietal	C		O	Pea
IV	Free central	D		P	China rose

- (1) I-D-O, II-A-M, III-B-N, IV-C-P
 (2) I-A-N, II-B-M, III-C-P, IV-D-O
 (3) I-D-O, II-B-P, III-C-M, IV-A-N
 (4) I-C-O, II-D-P, III-A-M, IV-B-N
120. How many plants in the list given below have marginal placentation ?
 Mustard, Tulip, Asparagus, chili, Arhar, sunhemp, moong, soyabean, colchicum, onion, Tobacco.
- (1) Six (2) Three (3) Four (4) Five

121. **Statement-I** : A leaf is said to be simple, when its lamina is entire or when incised, the incision do not touch the midrib.

Statement-II : When the incision of the lamina reach up to the midrib breaking it into a number of leaflets, the leaf is called compound leaf.

- (1) Statement-I & Statement-II both are correct.
 (2) Statement-I is correct & Statement-II is incorrect.
 (3) Statement-I is incorrect & Statement-II is correct.
 (4) Both Statement-I & Statement-II is incorrect.
122. The fruits present in wheat and rice are called :-
 (1) Cypsella (2) Achene
 (3) Siliqua (4) Caryopsis

123. Tricarpellary syncarpous gynoecium is found in flowers of :

- (1) Lily family
 (2) Potato family
 (3) Pulses family
 (4) Cucurbitaceae

124. Vegetative propagation in jasmine occurs by

- (1) Runner (2) Sucker
 (3) Stolon (4) Offset

125. They are converted into tendrils for climbing as in _____ or into spines for defence as in _____

- (1) Peas & Cacti
 (2) Citrus & peas
 (3) Mustard & Primrose
 (4) Cacti & cucumber

126. The membranous system of grana is responsible for :-

- (1) Trapping light energy and formation of sugar
 (2) Trapping CO₂ and formation of ATP
 (3) Trapping light energy and formation of ATP
 (4) Use glucose for the synthesis of ATP

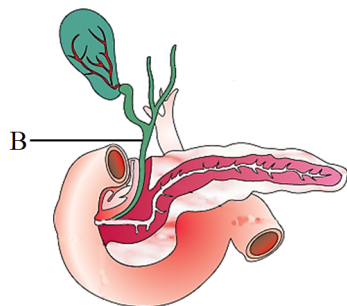
127. The reaction which is not directly light driven but dependent upon the product of light reaction (ATP & NADPH) is called as
- Dark reaction
 - Carbon reaction
 - Both (1) & (2)
 - Photochemical reaction
128. Synthesis of complex organic substances from simple inorganic raw materials in the presence of sunlight and chlorophyll is called as ____, which is a ____ process.
- Photosynthesis, anabolic
 - Photosynthesis, catabolic
 - Respiration, anabolic
 - Respiration, catabolic
129. All of the following are required for photosynthesis to occur, except :-
- CO₂
 - Light
 - Green pigment of the leaf
 - Starch
130. Refer the given reaction :-

$$2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$$
 where does this reaction take place in chloroplast of plant ?
- Outer surface of thylakoid membrane
 - Inner surface of thylakoid membrane
 - Stroma
 - Intermembrane space
131. In PS-I the reaction centre Chl a has absorption maxima at ____, while in PS-II the reaction centre Chl a has absorption maxima at
- P₆₈₀ - P₇₀₀
 - P₇₀₀ - P₆₈₀
 - P₈₀₀ - P₆₀₀
 - P₇₀₀ - P₉₀₀
132. Absorption spectrum of chl.a show maximum absorption in and regions of light:-
- Blue & Green
 - Blue & Red
 - Red & Green
 - Red & Far red
133. Action spectrum of photosynthesis resembles roughly the
- Absorption spectrum of chlorophyll a
 - Absorption spectrum of chlorophyll b
 - Absorption spectrum of chlorophyll c
 - Absorption spectrum of chlorophyll a and b
134. photochemical phase does not include
- light absorption
 - Water splitting and O₂ release
 - ATP and NADPH formation
 - CO₂ fixation
135. Which of the following is not a product of light reaction of photosynthesis ?
- Oxygen
 - NADPH
 - NADH
 - ATP

SECTION-B (BIOLOGY-I)

136. Hepato-pancreatic duct opens into the duodenum and carries
- Only bile
 - Only pancreatic juice
 - Both bile and pancreatic juice
 - Saliva
137. The back flow of faecal matter from the large intestine is prevented by the presence of :-
- Epiglottis
 - Sphincter of Oddi
 - Ileo-Caecal valve
 - Gastro-Oesophageal sphincter

138.



Identify 'B' in the above figure :-

- (1) Common Bile duct
- (2) Common Hepatic duct
- (3) Hepato-pancreatic duct
- (4) Duct from gall bladder

139. Which of the following statement is correct :-

- (a) Human digestive system consists of alimentary canal only
- (b) Oral cavity has a number of teeth & muscular tongue
- (c) Mouth leads to pharynx
- (d) Humans have incomplete alimentary canal

- (1) a & b
- (2) b & c
- (3) b only
- (4) a, b and c

140. Which of the following statements is/are incorrect regarding digestion and absorption of food in human beings :-

- (a) About 90% of starch is hydrolysed by salivary amylase in our oral cavity.
- (b) Mucus cells in our stomach secrete the proenzyme trypsinogen.
- (c) Vitamin-D is produced in human body in skin.
- (d) Bile salts act as activator of pancreatic lipase.

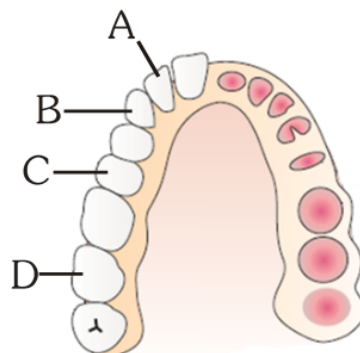
- (1) Two, a and b
- (2) Two, a and c
- (3) Two, a and d
- (4) Three, a, b and d

TG: @Chalnaayaaar

PHASE - MEF

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141. Identify A, B, C and D and choose correct option regarding their number in upper jaw:-



	A	B	C	D
(1)	Incisors-2	Canine-2	Premolars-2	Molars-3
(2)	Incisors-4	Canine-4	Premolars-8	Molars-12
(3)	Incisors-4	Canine-2	Premolars-4	Molars-6
(4)	Incisors-2	Canine-1	Premolars-2	Molars-3

142. Which statements are correct?

- (A) In cymose type of inflorescence, the main axis terminates in a flower, hence is unlimited in growth.
- (B) A typical flower has four different kinds of whorls arranged successively on the swollen end of the stalk, called pedicel.
- (C) When a flower can be divided into two similar halves only in one particular vertical plane, it is zygomorphic.
- (D) Flowers with bracts-reduced leaf found at the base of the pedicel are called bracteate.

- (1) All are correct
- (2) A and D
- (3) B and D
- (4) Only C and D

143. How many plant from the list given below belongs to family liliaceae ?

Lupin, Tulip, Petunia, Tobacco, Trifolium, Sesbania, Asparagus, Mulaithi, Aloe, Gloriosa

- (1) Four
- (2) Five
- (3) Six
- (4) Three

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144. Statement-I : A lateral branch with short internodes and each node bearing a rosette of leaves and tuft of roots is found in aquatic plants like *Pistia* and *Eichhornia*.

Statement-II : Banana, pineapple and *Chrysanthemum* are examples of sucker.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II is incorrect.

145. Which of the following plant possesses the placentation, ovules is not borne on peripheral wall ?

- (1) Cucumber
- (2) Argemone
- (3) Mustard
- (4) Primrose

146. Chlorophyll a appears ----- in colour & chlorophyll b appears ----- in colour on chromatogram :-

- (1) Blue green, Yellow green
- (2) Yellow green, Blue green
- (3) Blue, Red
- (4) Green, Red

147. Separation of leaf pigments of any green plants can be carried out by

- (1) X-ray radiography
- (2) Paper chromatography
- (3) Half-leaf experiment
- (4) Variegated leaf experiment

148. Stroma lamellae membrane lacks :-

- (1) PS I only
- (2) PS II only
- (3) PS I & electron carriers
- (4) PS II and NADP reductase enzyme

149. Identify the correct match from the column I, II and III :-

Column-I		Column-II		Column-III	
1	Predation	a	Plant pollinator Relation	i	+/-
2	Commensalism	b	Cuckoo & Crow	ii	+0
3	Mutualism	c	Biological control method	iii	+/+
4	Parasitism	d	Sea-anemone Clawn fish	iv	+/-

- (1) 1-b-i, 2-a-ii, 3-c-iii, 4-d-iv
- (2) 4-a-ii, 2-b-iii, 1-c-iv, 3-d-i
- (3) 1-c-i, 3-a-iii, 2-d-ii, 4-b-iv
- (4) 1-c-i, 3-a-iii, 2-b-iv, 4-d-ii

150. The correct statement for parasites is/are :-

- (a) host specific parasites & hosts tend to co-evolve
 - (b) parasites have highly developed sense organs
 - (c) parasites may reduce population density of host
 - (d) parasites have highly developed digestive system
- (1) a and b
 - (2) b and c
 - (3) a and c
 - (4) a and d

Topic : MONERA TO GYMNOSPERMS, MORPHOLOGY (COMPLETE), UPTO DIGESTIVE GLANDS (CHAPTER-DIGESTIVE SYSTEM), PHOTOSYNTHESIS-INTRODUCTION [BACK UNIT] : ECOLOGY

SECTION-A (BIOLOGY-II)

151. Barnacles growing on the back of whale is an example of :-

- (1) Mutualism (2) Parasitism
(3) Amensalism (4) Commensalism

152. When an important predator starfish *Pisaster* was experimentally removed from an enclosed intertidal area of communities of the American pacific coast, it resulted in extinction of more than 10 species of invertebrates with in a year. It demonstrates role of a predator as a factor for :-

- (1) preventing mutualism among prey species
(2) reducing the intensity of competition among competing prey species
(3) Over-exploitation of prey species
(4) reducing feeding efficiency in prey species

153. Which one is not a detrimental interaction ?

- (1) Predation (2) Amensalism
(3) Competition (4) Commensalism

154. The prickly pear cactus introduced in to Australia in the early 1920s caused havoc by spreading rapidly in to millions of hectares of rangeland because :-

- (1) There was no competition
(2) Due to less interspecific competition
(3) Absence of natural predator
(4) Availability of high nutrient

155. Coevolution, Coextinction, Coexistence can be seen in :-

- (1) Mutualism (2) Commensalism
(3) Protocooperation (4) Competition

156. **Statement-I** : CNG is better than diesel.

Statement-II : Unleaded & low sulphur petrol is less polluting.

- (1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct.

157. During succession :-

- (1) Species diversity and biomass increase, but humus decreases
(2) Species diversity and humus increase, but biomass decreases
(3) Humus and biomass increase, but species diversity decreases
(4) Species diversity, humus and biomass increase

158. Which of the following is an example of colloidal water pollutant ?

- (1) Ammonia (2) Bacteria
(3) Sand (4) Phosphate

159. If a single plant species is removed from a food web then most likely :-

- (1) Dependent herbivores will have to find new food resources
(2) Carnivores will be eliminated from that zone
(3) Other plants will produce enough food for herbivores
(4) An animal species will fill the unoccupied space

160. The group of plant kingdom in which vegetative reproduction takes place by fragmentation and buds on protonema and sex organs are produced at the apex of leafy shoot and after fertilization zygote develops into sporophyte via embryo formation is ?

- (1) Liverworts (2) Mosses
(3) Ferns (4) Lycopodes

161. Which of the following is true for alternation of generation ?

- (1) The sporophyte undergoes syngamy to produce spores
(2) The gametophyte undergoes syngamy to produce spores
(3) The sporophyte undergoes meiosis to produce spores
(4) The gametophyte undergoes meiosis to produce gametes

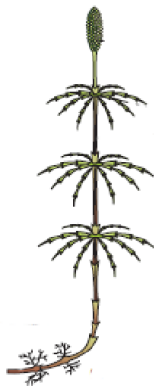
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162. Identify given figure and correct information about it :-



- (1) Pteridophyte, Gametophyte of *Equisetum*
 (2) Pteridophyte, Sporophyte of *Equisetum*
 (3) Bryophyte, Gametophyte of *Funaria*
 (4) Pteridophyte, Gametophyte of *Salvinia*.
163. The sporophyte in mosses is more elaborate as compared to which of following plants groups ?
 (1) *Polysiphonia* (2) *Marchantia*
 (3) *Ectocarpus* (4) All of these
164. Choose the incorrect statement of the following:-
 (1) Compared to many other organisms, bacteria as a group show the most extensive metabolic diversity.
 (2) Exceptionally some bacteria have small extra circular DNA with histones
 (3) Mesosomes are infoldings of cell membrane
 (4) Some bacteria have photosynthetic ability also
165. Which of the following bacteria are responsible for the production of biogas from the dung of cows and buffaloes?
 (1) Thermoacidophiles (2) Halophiles
 (3) Methanogens (4) Cyanobacteria
166. The statement '*contagium vivum fluidum*' was given by :-
 (1) Ivanowsky (2) M.W. Beijerinck
 (3) Stanley (4) Pasteur

167. Match the classes of fungi given in Column I with their common names given in Column II :

	Column I		Column II
(i)	Phycomycetes	(a)	Imperfect fungi
(ii)	Ascomycetes	(b)	Club fungi
(iii)	Basidiomycetes	(c)	Sac fungi
(iv)	Deuteromycetes	(d)	Algal fungi

Options :

- (1) (i)-a (ii)-b (iii)-c (iv)-d
 (2) (i)-d (ii)-c (iii)-b (iv)-a
 (3) (i)-c (ii)-b (iii)-d (iv)-a
 (4) (i)-b (ii)-a (iii)-c (iv)-d
168. Which of the following character is similar in both gametophyte and sporophyte of both moss and fern?
 (1) Thalloid body
 (2) Presence of vascular tissue
 (3) Photosynthetic nature
 (4) Free living existence
169. Agar and carrageen like hydrocolloids are obtained from :
 (1) Green algae (2) Red algae
 (3) Brown algae (4) Blue green algae
170. Capsule is advantageous to bacteria because it:
 (1) Hides the bacteria from host immune system
 (2) Prevents the bacteria from desiccation
 (3) Provide attachment to host tissue
 (4) Produces antibodies
171. **Statement-I** : Boundaries of kingdom Protista are not well defined.
Statement-II : Kingdom protista includes plant like animal-like and fungi-like organisms.
 (1) Statement-I & Statement-II both are correct.
 (2) Statement-I is correct & Statement-II is incorrect.
 (3) Statement-I is incorrect & Statement-II is correct.
 (4) Both Statement-I & Statement-II is incorrect.

172. Match the Column-I with Column-II

	Column-I		Column-II
(A)	Sac fungi	(i)	<i>Aspergillus</i>
(B)	Algal fungi	(ii)	<i>Alternaria</i>
(C)	Club fungi	(iii)	<i>Mucor</i>
(D)	Imperfect fungi	(iv)	<i>Puccinia</i>

- (1) A-i, B-iii, C-iv, D-ii
 (2) A-i, B-ii, C-iii, D-iv
 (3) A-ii, B-i, C-iv, D-iii
 (4) A-iii, B-i, C-ii, D-iv

173. What is true for virus?

- (1) Plant virus generally has ds DNA
 (2) Virus use host ribosomes to synthesize viral proteins
 (3) Bacteriophage can infect other virus.
 (4) A virus contain both DNA and RNA

174. Which of the following structure provide attachment of bacteria to rocks and host tissues?

- (1) Glycocalyx (2) Plasmid
 (3) Fimbriae (4) Flagella

175. Prothallus is:

- (1) Free-living sporophyte of moss
 (2) Free-living gametophyte of moss
 (3) Free-living sporophyte of fern
 (4) Free-living gametophyte of fern

176. **Assertion** : Endosperm in gymnosperm is haploid.

Reason : Endosperm in gymnosperm is multicellular

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but Reason is False
 (4) Assertion is False but Reason is True

177. Which of the following method is for ex-situ conservation?

- (1) Botanical gardens (2) National parks
 (3) Wildlife sanctuaries (4) Hot spots

178. Humus is

- (1) Amorphous substances
 (2) Colloidal nature
 (3) Highly resistant to microbial action
 (4) All of the above

179. Population does not has following character?

- (1) Dominance (2) Birth rate
 (3) Death rate (4) Sex-ratio

180. Which device removes 99% particulate matter from thermal power plant exhaust ?

- (1) Electrostatic precipitator
 (2) Wet scrubber
 (3) Catalytic converters
 (4) Dry scrubber

181. Which green-house gas maximum contribute global warming?

- (1) CFC (2) N₂O (3) CO₂ (4) CH₄

182. The interaction is detrimental to both the species, is

- (1) Predation (2) Commensalism
 (3) Amensalism (4) Competition

183. How much portion of the PAR is captured by plants?

- (1) 5–10% (2) 7–10%
 (3) 8–10% (4) 2–10%

184. Which of the following level of sound may damage ear drum and can impair the hearing ability permanently?

- (1) 80 dB (2) 100 dB
 (3) 120 dB (4) 150 dB or more

185. In clean water :

- (1) BOD is low but DO is high
- (2) Both BOD and DO are high
- (3) BOD is high but DO is low
- (4) Both BOD and DO are low

SECTION-B (BIOLOGY-II)

186. Read the statement A–D :-

- (A) The community that is near equilibrium with the environment is called climax community
- (B) The gradual and fairly predictable change in the species composition of a given area called ecological succession.
- (C) The species that invade a bare area called pioneer species.
- (D) Hydrarch succession takes place in dry areas.

How many above statements are correct ?

- (1) Two (2) Three (3) Four (4) One

187. Temperature affects → "A" → Basal metabolism, activity and other physiological functions of the organism.

What is "A" with respect to the given significance of temperature ?

- (1) Life span (2) Range of tolerance
- (3) Enzyme kinetics (4) All of these

188. Male and female gametophyte do not have free-living existence in :-

- (1) *Lycopodium* (2) *Equisetum*
- (3) *Salvinia* (4) *Polytrichum*

189. Absence of nuclear membrane and presence of non-cellulosic cell wall is character of :-

- (1) Monera & Protista
- (2) Only monera
- (3) Fungi and Animalia
- (4) Protista & Monera

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190. Choose the incorrect statement of following :-

- (1) Virus show mutation
- (2) Virus are specific towards host
- (3) Virus can multiply without host cell
- (4) Virus are obligate parasite

191. Dikaryophase of fungus is observed in :

- (1) Phocomycetes and ascomycetes
- (2) Phycomycetes and basidiomycetes
- (3) Phycomycetes, ascomycetes and basidiomycetes
- (4) Basidiomycetes and ascomycetes

192. Select the incorrect statement in the following:-

- (1) Chrysophytes are found in fresh water as well as in marine environment.
- (2) Dinoflagellates are mostly fresh water organism & photosynthetic
- (3) Majority of Euglenoids are fresh water organism
- (4) Slime moulds are saprophytic protist

193. Isogamy with non-flagellated gametes shown by :

- (1) *Fucus* (2) *Volvox*
- (3) *Spirogyra* (4) *Eudorina*

194. Fruits are not formed in gymnosperms due to absence of:

- (1) Vessels (2) Ovules
- (3) Ovary (4) Archegonia

195. **Assertion** : Low temperature anaerobiosis and low soil moisture inhibit decomposition.

Reason : Above conditions reduce growth of most of bacteria and fungi.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false.
- (3) **Assertion** is false but **Reason** is true.
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

196. Membranous extensions in bacteria containing oxidizing enzymes are known as :
- (1) Chromatophores (2) Mesosomes
(3) Inclusion bodies (4) Plasmid
197. Which of the following is not included in evil quartet of biodiversity loss?
- (1) Habitat loss and fragmentation
(2) Over-exploitation
(3) Global warming
(4) Alien species invasions
198. Which of the following is not a broadly utilitarian?
- (1) Oxygen production
(2) Pollination
(3) Mineral recycling
(4) Medicines production

199. When resources are un-limited then population show:
- (1) Logistic growth
(2) Exponential growth
(3) S-shaped growth curve
(4) Both (1) and (3)
200. The amount of organic matter in sewage water can be estimated by :-
- (1) Measuring oxygen demand
(2) Weight of microorganisms
(3) Salt analysis
(4) Calorimetry

 ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)		TOPICS TO BE COVERED IN NEXT ASSIGNMENT
		DATE : 08-10-2023 (Session : 2023 - 2024)
ENTHUSE ADV. COURSE : MEF		
P	THERMAL PHYSICS AND CAPACITOR	
C	GOC - II	
B	GENETICS (MOD. 09 & 10)	